Algebra 7 — Solving quadratic equations · Quadratic inequalities

Calculator allowed. Marks in brackets. ★ = harder question. Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

1 Solve by factorising. [5]

a $3x^2 - 48 = 0$ **b** $2x^2 + 5x - 3 = 0$ **c** $x(3x - 7) = 6$

a $3(x^2 - 16) = 0 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. **b** $(2x - 1)(x + 3) = 0 \Rightarrow x = \frac{1}{2}$ or $x = -3$. **c** Expand and bring everything to one side first: $3x^2 - 7x - 6 = 0 \Rightarrow (3x + 2)(x - 3) = 0 \Rightarrow x = -\frac{2}{3}$ or $x = 3$.

2 **a** Complete the square for $x^2 - 6x + 4$. **b** Hence solve $x^2 - 6x + 4 = 0$, giving your answers in surd form. [4]

a $x^2 - 6x + 4 = (x - 3)^2 - 9 + 4 = (x - 3)^2 - 5$. **b** $(x - 3)^2 = 5 \Rightarrow x - 3 = \pm\sqrt{5} \Rightarrow x = 3 + \sqrt{5}$ or $x = 3 - \sqrt{5}$.

3 Use the quadratic formula to solve, giving answers in surd form. [4]

a $x^2 + 3x - 2 = 0$ **b** $3x^2 - 5x + 1 = 0$

a $a = 1, b = 3, c = -2$: $x = \dfrac{-3 \pm \sqrt{9 + 8}}{2} = \dfrac{-3 \pm \sqrt{17}}{2}$. **b** $a = 3, b = -5, c = 1$: $x = \dfrac{5 \pm \sqrt{25 - 12}}{6} = \dfrac{5 \pm \sqrt{13}}{6}$.

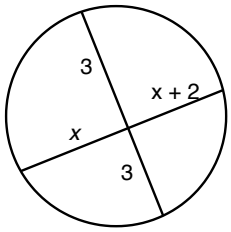
4 Solve each quadratic inequality and show the solution set on a number line. [4]

a $3x^2 > 27$ **b** $x^2 - 2x - 8 \leq 0$

Sketch the parabola and read off where it is above or below the x-axis.

a $x^2 > 9$: the graph of $y = x^2 - 9$ is above the axis outside the roots ± 3, so $x < -3$ or $x > 3$ — two separate parts (open circles at -3 and 3, arrows outwards). **b** $(x - 4)(x + 2) \leq 0$: roots -2 and 4; the U-shaped graph is below (or on) the axis between them, so $-2 \leq x \leq 4$ — one part (filled circles, line between).

5 Two chords of a circle cross as shown. **a** Use the intersecting chords theorem to form an equation in x. **b** Solve it to find x, correct to 3 significant figures. [4]



a $x(x+2) = 3 \times 3 \Rightarrow x^2 + 2x - 9 = 0$. **b** $x = \dfrac{-2 \pm \sqrt{4+36}}{2} = -1 \pm \sqrt{10}$. A length is positive, so $x = -1 + \sqrt{10} = 2.16$ (3 s.f.).

6 A rectangular poster is 4 cm longer than it is wide. Its area is 60 cm². Find its width. [4]

Width w, length $w+4$:
$w(w+4) = 60 \Rightarrow w^2 + 4w - 60 = 0 \Rightarrow (w+10)(w-6) = 0$. $w = -10$ is impossible, so the width is 6 cm (and the length 10 cm; check $6 \times 10 = 60$ ✓).

Total: 25 marks

Check: bring everything to one side (= 0) before factorising · completing the square: $(x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$ · formula: write a, b, c with signs, b^2 is positive, $-4ac$ positive if a or c is negative · inequality: sketch, roots, "below" = between, "above" = outside · reject negative lengths.